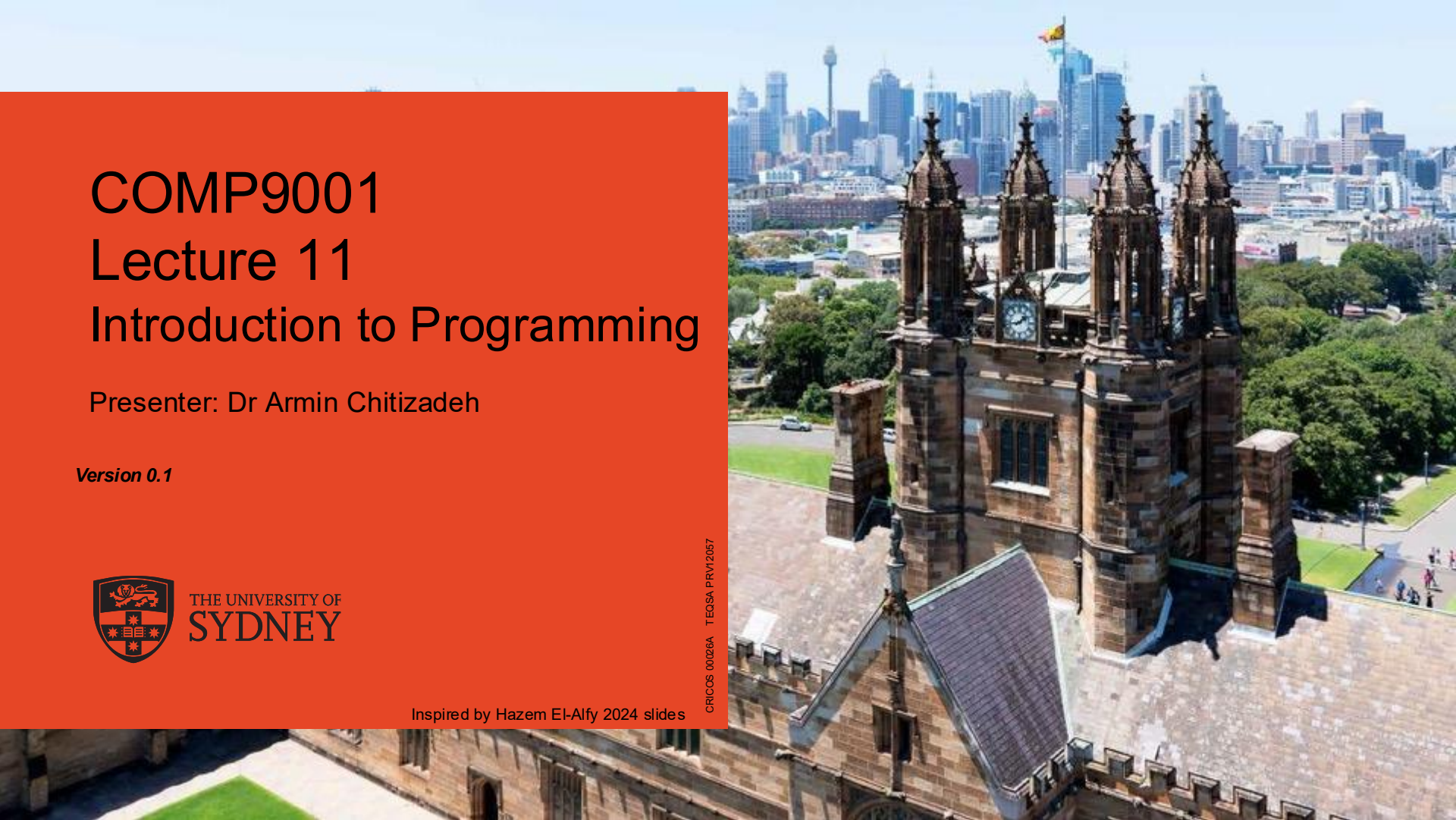
COMP9001
Lecture 11
Introduction to Programming

Presenter: Dr Armin Chitizadeh

Version 0.1



THE UNIVERSITY OF SYDNEY

Inspired by Hazem El-Alfy 2024 slides

We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

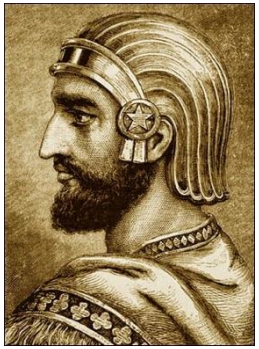
THE UNIVERSITY OF SYDNEY

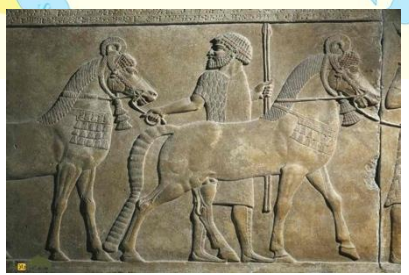
Recursion

Why are We Learning **Recursion** Technique?

- Many problems that can be done with recursion can be done with iteration
    - But for some, recursion can be way easier
- It is a common technique in AI
- It differentiate great programmer against normal programmer
- With some modifications, it can be extremely efficient (will be covered in other units)

Chapar Khaneh (Postal Service System)





Chapar Khaneh (Postal)

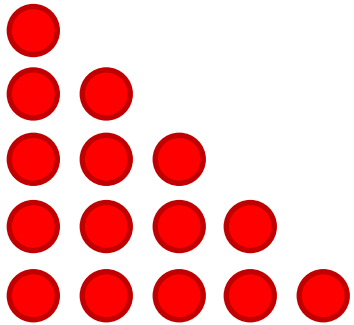
Chapar Khaneh (Postal)

Let’s summarise what you saw on the last slide:

- Each postman had to wait and be paid for the next postman to arrive
- As soon as they returned the information, their contract was finished
- Each postman, called another postman
- They used what the next postman said to generate their answer

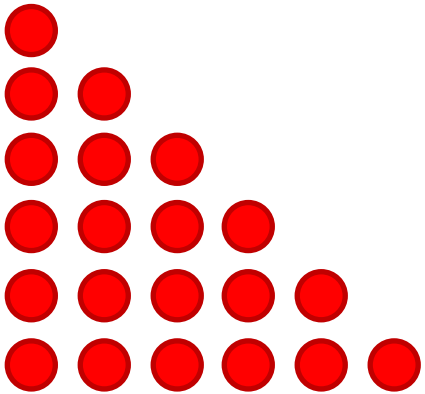
Calculating the sum of the first n natural numbers

(n = 5)



Calculating the sum of the first n natural numbers

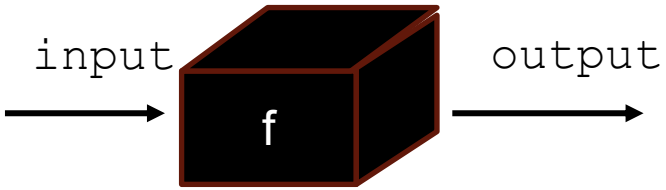
(n = 6)



$$sum(6) = sum(5) + 6$$

$$sum(n) = sum(n-1) + n$$

Remember Functions?



$$sum(n) = \begin{cases} \\ sum(n\text{-}1) + n \end{cases}$$

What is Recursion?

- A function that calls itself is **recursive.**
- The process of executing is called **recursion**.

What is Recursion?

- Whenever you solve a **big problem** by cheating and using the solution to a **smaller problem** to get to the solution to your **big problem.**

  - *Professor Richard Buckland*

- Recursion is a problem-solving technique
  - Solving a problem by solving smaller version of itself

What is Recursion?

Demo!

Calculating the sum of the first n natural numbers

$$sum(6) = sum(5) + 6$$

$$sum(n) = sum(n-1) + n$$

Recursion Parts

```python
######
## Calculating the sum of the first n natural numbers
######
######

def sum_natural(n):
    if n == 1:
        return 1
    else:
        print(f"n:{n}")
        answer = sum_natural(n-1) + n
        return answer


print(f"sum_natural(6): {sum_natural(6)}")
```

- **Base case:**
  - The simplest case
    - •Also know as stopping condition

- **Recursion relation**:
  - a mathematical equation that defines a sequence by relating each term to one or more previous terms in the sequence
  - In each recursion relation there is a recursive call:
    - •Recursive call is when a function calls the same function but with different arguments

Recursive Call

```
######
## Calculating the sum of the first n natural numbers
######
######

def sum_natural(n):
    if n == 1:
        return 1
    else:
        answer = sum_natural(n-1) + n
        return answer


print(f"sum_natural(3): {sum_natural(3)}")
```

```
def sum_natural(3):
    if 3 == 1:
        return 1
    else:
        answer = sum_natural(3-1) + 3
        return answer
```

```
def sum_natural(2):
    if 2 == 1:
        return 1
    else:
        answer = sum_natural(2-1) + 2
        return answer
```

```
def sum_natural(1):
    if 1 == 1:
        return 1
    else:
        answer = sum_natural(1-1) + 1
        return answer
```

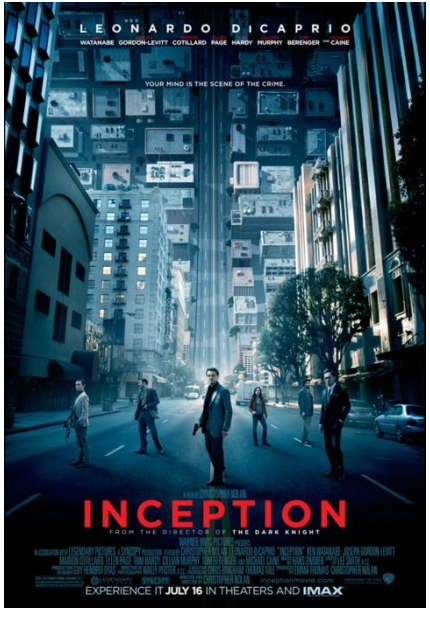
Different Ways to Implement the Base Case in Python

```
######
## Calculating the sum of the first n natural numbers
######
######

def sum_natural(n):
    if n == 1:
        return 1
    else:
        answer = sum_natural(n-1) + n
        return answer


print(f"sum_natural(3): {sum_natural(3)}")
```

Different Ways to Implement the Base Case in Python



Different Ways to Implement the Base Case in Python

- Let’s write different variants of base-case for sum_natural recursive function

More Examples
Counter

- Using the recursion write a program that prints from n to 1

- Can you make it print in reverse with minimal change?

One common mistake

Infinite Recursion

Infinite Recursion

- Sequence of calls must reach its base case, otherwise it goes on making recursive calls forever (*infinite recursion*)

Recursive Call

```
######
## Calculating the sum of the first n natural numbers
######
######

def sum_natural(n):
    if n == 1:
        return 1
    else:
        answer = sum_natural(n-1) + n
        return answer


print(f"sum_natural(3): {sum_natural(3)}")
```

```
def sum_natural(3):
    if 3 == 1:
        return 1
    else:
        answer = sum_natural(3-1) + 3
        return answer
```

```
def sum_natural(2):
    if 2 == 1:
        return 1
    else:
        answer = sum_natural(2-1) + 2
        return answer
```

```
def sum_natural(1):
    if 1 == 1:
        return 1
    else:
        answer = sum_natural(1-1) + 1
        return answer
```

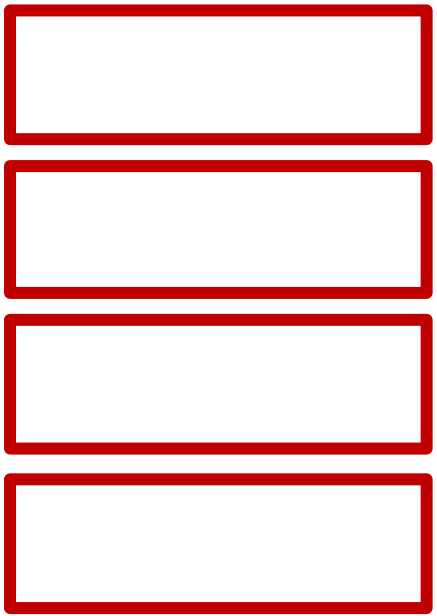
Issue with Recursive Calling memory usage

Issue with Recursive Calling memory usage

```python
######
## Calculating the sum of the first n natural numbers
######
######

def sum_natural(n):
    if n == 1:
        return 1
    else:
        answer = sum_natural(n-1) + n
        return answer


print(f"sum_natural(3): {sum_natural(3)}")
```

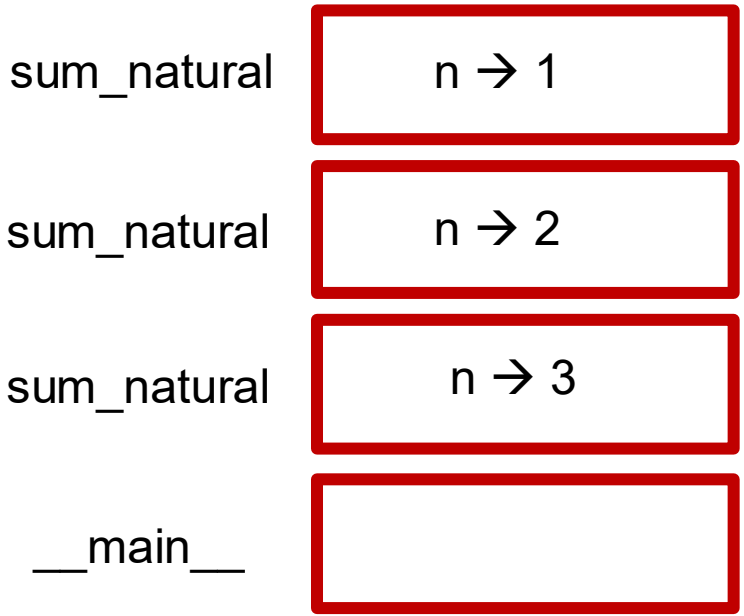


Issue with Recursive Calling memory usage

```python
######
## Calculating the sum of the first n natural numbers
######
######

def sum_natural(n):
    if n == 1:
        return 1
    else:
        answer = sum_natural(n-1) + n
        return answer


print(f"sum_natural(3): {sum_natural(3)}")
```



So Why Are We Using Recursion?

Let’s do an experiment!

- Finding the max value in a binary tree
- Vs
- Finding the max number in a list

- We need volunteers:
    - 15 volunteers to return the max number, given 3 numbers
    - 1 volunteer to return the max number, given 15 numbers
    - 1 volunteer to live stream the event to echo360. (*Tools provided*)

So Why Are We Using Recursion?

Binary Tree Max Finder

```python
class TreeNode:
    def __init__(self, value, left=None, right=None):
        self.value = value
        self.left = left
        self.right = right

def find_max(node):
    if node is None:
        return -100

    left_max = find_max(node.left)
    right_max = find_max(node.right)

    return max(node.value, left_max, right_max)

```

```python
if __name__ == "__main__":
    # These structures will not be asked in the exam
    # Constructing a full binary tree with 15 nodes
    # Tree structure:
    #                10
    #              /    \
    #            32       14
    #          /  \       /  \
    #        53   78     12    66
    #       / \  / \    / \   / \
    #      21 41 71 92 81 93 75 17

    root = TreeNode(10,
        left=TreeNode(32,
            left=TreeNode(53,
                left=TreeNode(21),
                right=TreeNode(41)),
            right=TreeNode(78,
                left=TreeNode(71),
                right=TreeNode(92))),
        right=TreeNode(14,
            left=TreeNode(12,
                left=TreeNode(81),
                right=TreeNode(93)),
            right=TreeNode(66,
                left=TreeNode(75),
                right=TreeNode(17)))
    )

    max_value = find_max(root)
    print("Maximum value in the tree:", max_value)
    # Can you tell in which orders nodes are visited?
```

5 slow motorcycles vs 1 fast car
Who would win?

More Examples

Calculate the sum of items in a list using recursion

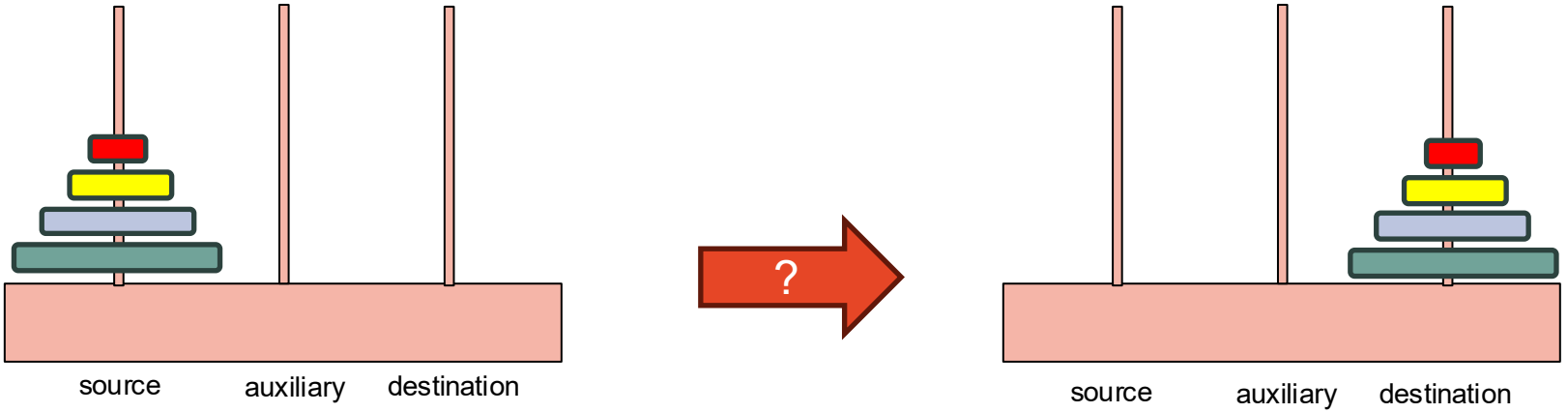
More Recursion Examples
Tower of Hanoi

Etsy Link

Let's Play the Game!

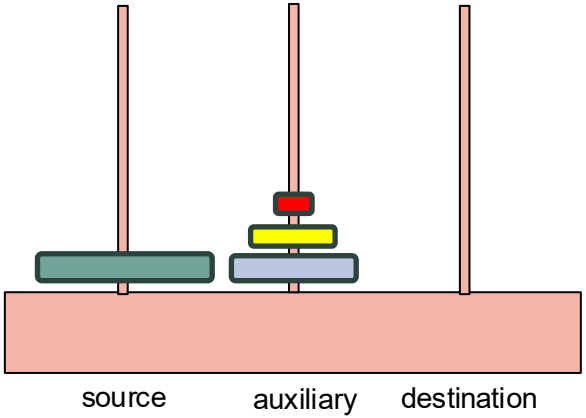
https://www.mathsisfun.com/games/towerofhanoi.html

More Recursion Examples
Tower of Hanoi

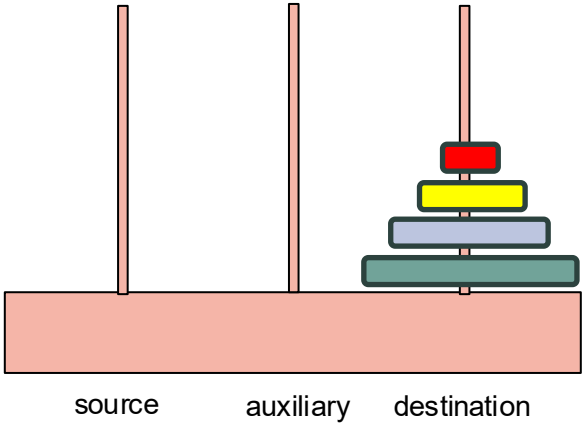


The Tower of Hanoi puzzle involves moving a stack of disks from source rod to destination rod, following specific rules.
You can only move one disk at a time, and a larger disk cannot be placed on top of a smaller disk.
The goal is to move all disks from the starting rod to a destination rod, using an auxiliary rod as needed.

More Recursion Examples
Tower of Hanoi



EASY



More Recursion Examples
You Choose!

Geeks for Geeks Recursion Interview Questions

- https://www.geeksforgeeks.org/top-50-interview-problems-on-recursion-algorithm/

- Vote!
- https://www.menti.com/alne5ocvsy61

Thanks you!

https://forms.office.com/r/4Q0die7fFK

Do you know what this is?